

**Reference**

**Modern electrical  
machines**

# Software Information

# Reference

## Equipment: modes of use

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The Modern Electrical Machines equipment can be used in two ways:

### Manual mode - no software

When no USB lead is plugged into the unit then the control unit is in manual mode. This allows users control of the DC and AC power supply outputs using the potentiometers on the front of the unit. The capacitor start / run is controlled by the front panel switch and the speed of the dynamometer is displayed by the 7 segment display. Manual mode is useful when teaching students the basics of electrical machine control and will be useful when used with true RMS multimeters to measure voltages and currents.

In Manual mode we make use of a small balance to measure torque. The swinging arm of the Dynamometer has two modes of use: when rotating counter-clockwise it presses

down on the balance. When rotating clockwise it presses down on the load cell for computer based measurement. Users simply reverse the direction of motion of the machine under test to change rotational direction.

### Software mode

When the USB lead is plugged in to the Control unit the front panel controls are disabled. Six separate software applications are available to the user for carrying out experiments with the different machines.

# Reference

## Understanding the software

# Modern electrical machines



### Applications

Six different PC based applications are shipped with the unit. Each of them is a self contained executable '\*.BAT' file. Drivers for the Control unit will need installing - the applications will not. These applications are:

**Open Control\_DC.bat:** for direct DC measurement for DC motors.

**Open Log\_DC.bat:** for gathering speed torque data and logging the results for DC motors.

**Open Control\_1Phase.bat:** for direct measurement for single phase AC induction machines.

**Open Log\_1Phase.bat:** for gathering speed torque data and logging the results for single phase AC induction machines.

**Open Control\_3Phase.bat:** for direct measurement for three phase induction and brushless DC motors.

**Open Log\_3Phase.bat:** for gathering speed torque data and logging the results for three phase induction and brushless DC motors.

These are explained below.

An API for use with LabView or MATLAB is also available.

Naming of these applications may vary slightly as updates are released.

# Reference

## Software flags

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Each application has a set of software flags which are displayed on the top left hand side. These are:

**Application:** green when the application is in Play mode.

**USB:** Green when there is a USB connection between the control unit and the PC.

**Error:** red when there is an Error in the control unit. Most often this happens due to an over current error. The unit is capable of providing 100 watts of power. When this is exceeded the unit shuts down. In this case you should lower the output voltage and try again.

**Motor:** green when the microswitches in the Dynamometer are activated by the presence of a machine under test. The unit will not run unless a machine under test is present.

**Direction:** green when the direction of rotation

is clockwise which will press down on the torque sensor.

**Run:** green when the run switch is triggered.

# Reference

## Understanding the software

# Modern electrical machines



There are two types of software application supplied with the Electrical Machines course:

- Manual control
- Automatic

Manual control applications allow you to manually set all of the parameters inside the control box. Automatic applications allow you to automatically take multiple sets of readings for producing speed torque graphs. This page and the following 5 pages show you the screen layouts and controls.

The file name for this program is 'Open Control\_DC.bat' and it is intended for manual DC measurements.

- 1) Program control - hit 'PLAY' to start
- 2) Flags for App running, USB comms and Motor in place etc. The box will not start until all show green ticks.

- 3) DC1 output - shows PWM duty cycle for DC output 1 with voltage and current.

- 4) Run / Stop switch - needs pressing to turn outputs on

- 5) DC2 output - shows PWM duty cycle for DC output 2

- 6) Meter for Current terminals

- 7) Meter for Voltage terminals

- 8) Dynamometer load as a %

- 9) Dynamometer speed in RPM

- 10) Torque in mNm

- 11) Dynamometer power in watts

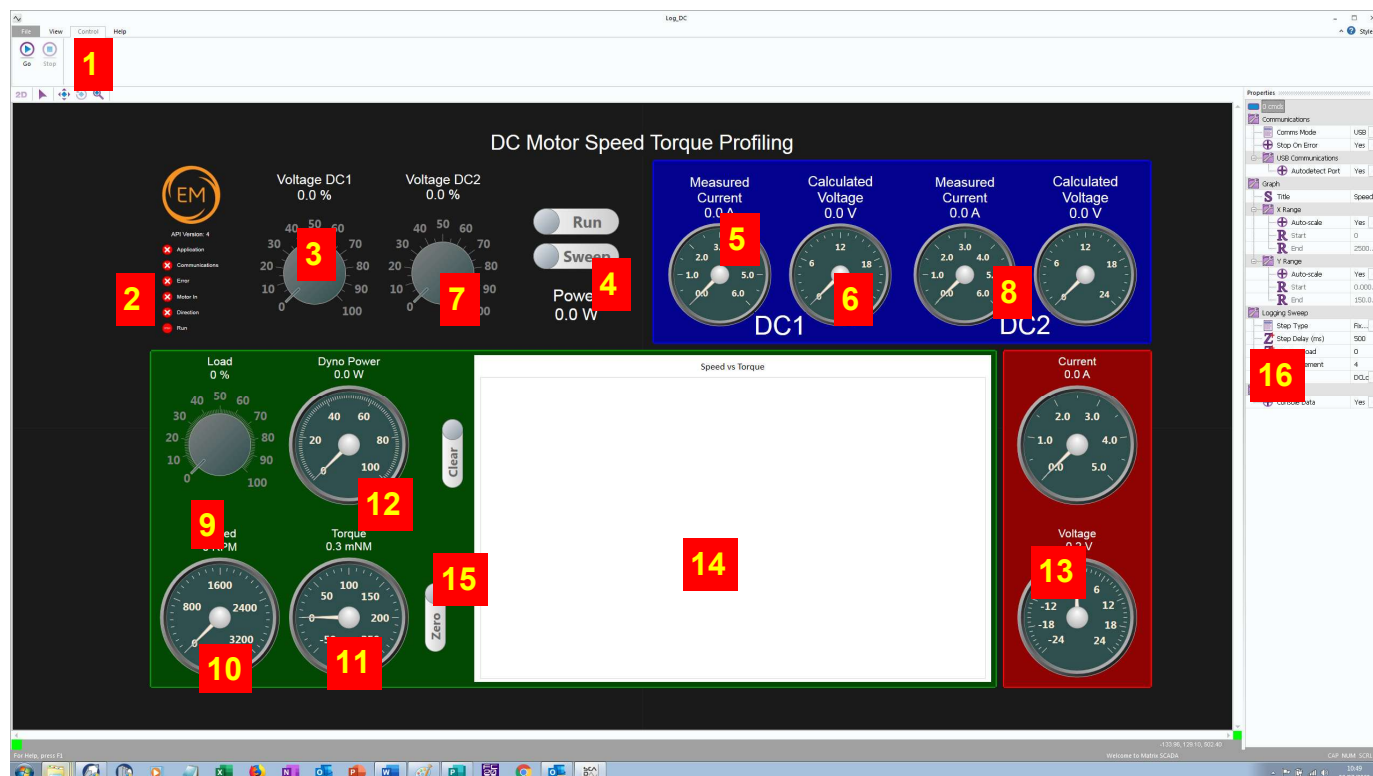
- 12) Oscilloscope

- 13) Properties for the program and scope.

# Reference

## Automatic DC software

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The file name for this program is 'Open Log\_DC.bat'

- 1) Program control - hit 'PLAY' to start
- 2) Flags for App running, USB comms and Motor in place. The box will not start until all three show green ticks
- 3) DC1 output - shows PWM duty cycle for DC output 1
- 4) Sweep - takes a set of readings sweeping the Dynamometer from 0-% to 100%
- 5) RUN and SWEEP switches.
- 6) DC2 output - shows PWM duty cycle for DC output 2
- 7) Meters for DC 1 - RMS values
- 8) Meters for DC 2 - RMS values
- 9) Dynamometer load as a %
- 10) Dynamometer speed in RPM

11) Torque in mNm

12) Dynamometer power in watts

13) Internal ammeter and voltmeter V1 and I1

14) On screen graph of plot data

15) Graph / log clear button

16) Program properties

When the Run button is pressed the control box will - at the specified voltage on DC1 and DC2 take readings from 0 load to 100% load in 2% increments. The results will be stored in the file LOG\_DC1.CSV for plotting in Excel.

Whilst the readings are being taken the on screen graph will show an approximate plot so that you can be sure you are getting valid data. Up to 8 data sets can be captured and displayed at one time.

# Reference

## Single phase AC manual

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The file name for this program is 'Open Control\_1Phase.bat'

- 1) Program control - hit 'PLAY' to start
- 2) Flags for App running, USB comms and Motor in place. The box will not start until all three show green ticks
- 3) Frequency of output
- 4) AC voltage output
- 5) SINE/DIGITAL: Sine wave or square wave output.
- 6) CAP START/CAP RUN: selects the internal capacitor value.
- 7) RUN switch - needs pressing to turn outputs on.
- 8) Meters for the outputs - RMS values
- 9) Dynamometer load as a %

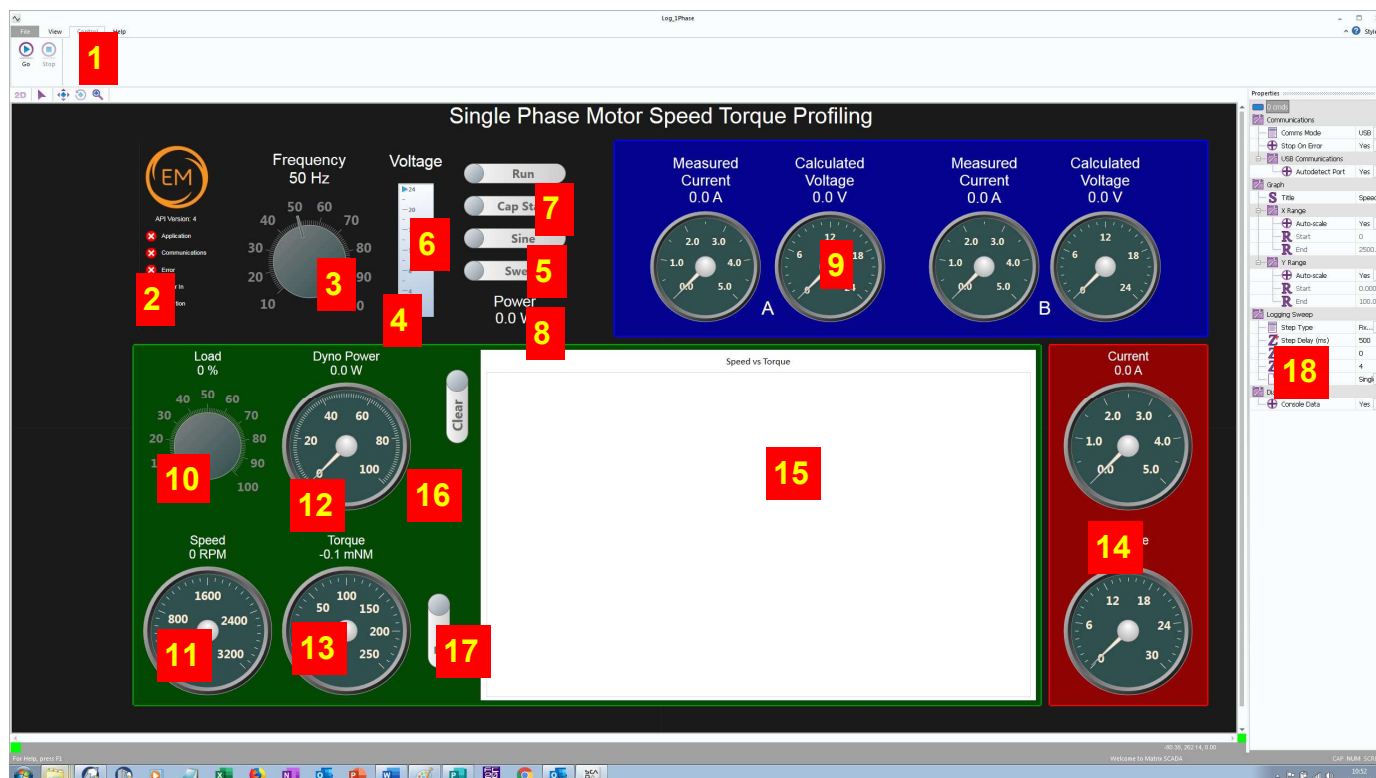
- 10) Dynamometer speed in RPM
- 11) Torque in mNm
- 12) Dynamometer power in watts
- 13) REAL V / EFFECTIVE V: shows the voltage on the graph in real terms or effective terms.
- 14) PHASE / TIME: shows X axis in degrees or in mS
- 15) CAPTURE: captures a waveform
- 16) Oscilloscope / graph plot area
- 17) Internal ammeter and voltmeter V1 and I1
- 18) Properties control



# Reference

## Single phase AC automatic

# Modern electrical machines



The file name for this program is 'Open Log\_1Phase.bat'

- 1) Program control - hit 'PLAY' to start
- 2) Flags for App running, USB comms and Motor in place. The box will not start until all three show green ticks
- 3) Frequency of output
- 4) Voltage of output
- 5) Run switch - starts automatic logging of data
- 6) Drivemode - default is pseudo-sine, on is trapezoid
- 7) CAP START / RUN switch controls the value of internal capacitor on the variable capacitor terminals
- 8) SWEEP: starts sweep function
- 9) Meters for the output - RMS values
- 10) Dynamometer load as a %

- 11) Dynamometer speed in RPM
- 12) Torque in mNm (Zero switch is used on start up)
- 13) Dynamometer power in watts
- 14) Internal ammeter and voltmeter V1 and I1
- 15) On screen graph of plot data
- 16) Graph / log clear button
- 17) Torque zero button
- 18) Properties

When the SWEEP button is pressed the control box will - at the specified frequency output take readings from 0 load to 100% load in 2% increments. The results will be stored in the file LOG\_1phase.CSV for plotting in Excel.

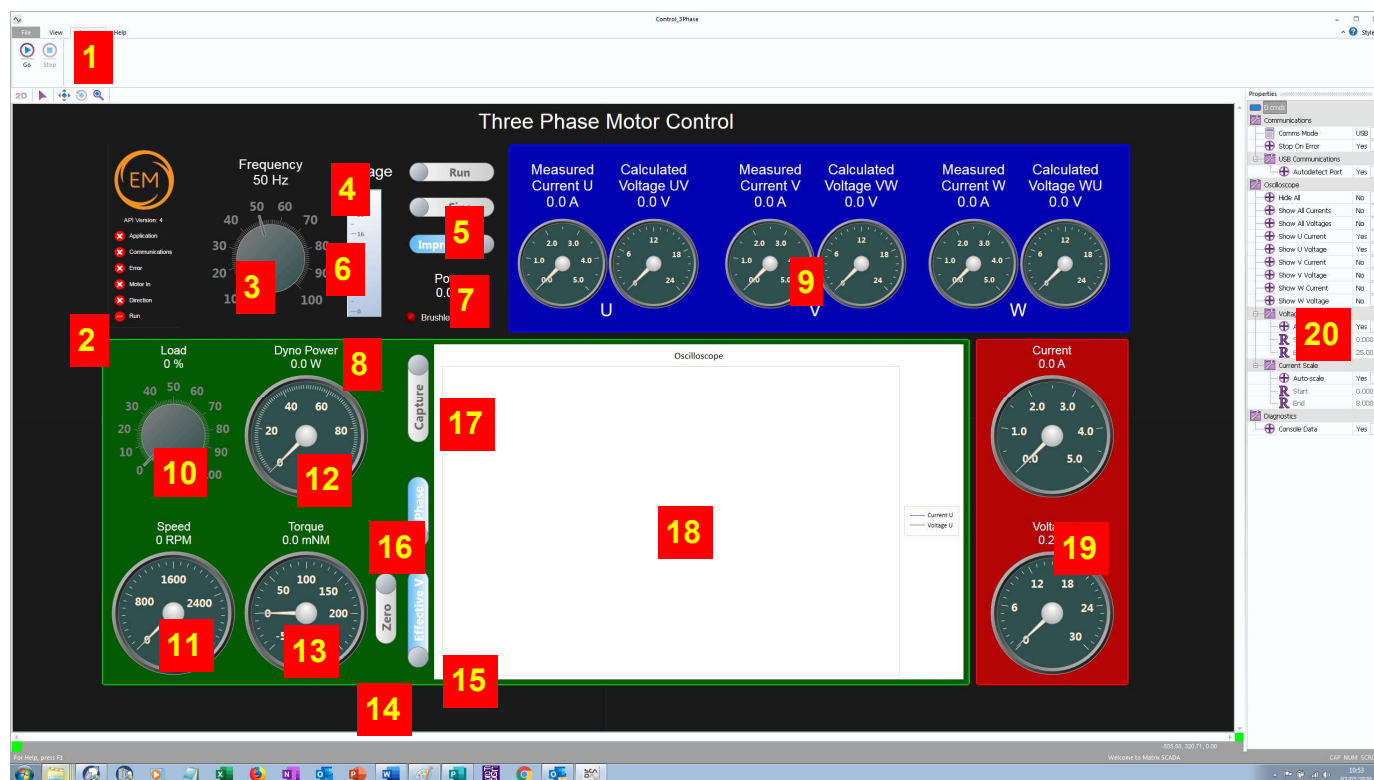
Whilst the readings are being taken the on screen graph will show an approximate plot so that you can be sure you are getting valid data.



# Reference

## Three phase AC manual

# Modern electrical machines



The file name for this program is 'Open Control\_3Phase.bat'

- 1) Program control - hit 'PLAY' to start
- 2) Flags for App running, USB comms and Motor in place.
- 3) Frequency of output
- 4) RUN: needs pressing to turn outputs on
- 5) SINE / DIGITAL: Sine or Square wave output
- 6) 20V / 12V output: alters the output voltage
- 7) IMPROVED / STANDARD: alters output waveform quality
- 8) BRUSHLESS MODE: indicates Brushless motor is plugged in
- 9) Meters for the output - RMS values
- 10) Dynamometer load as a %

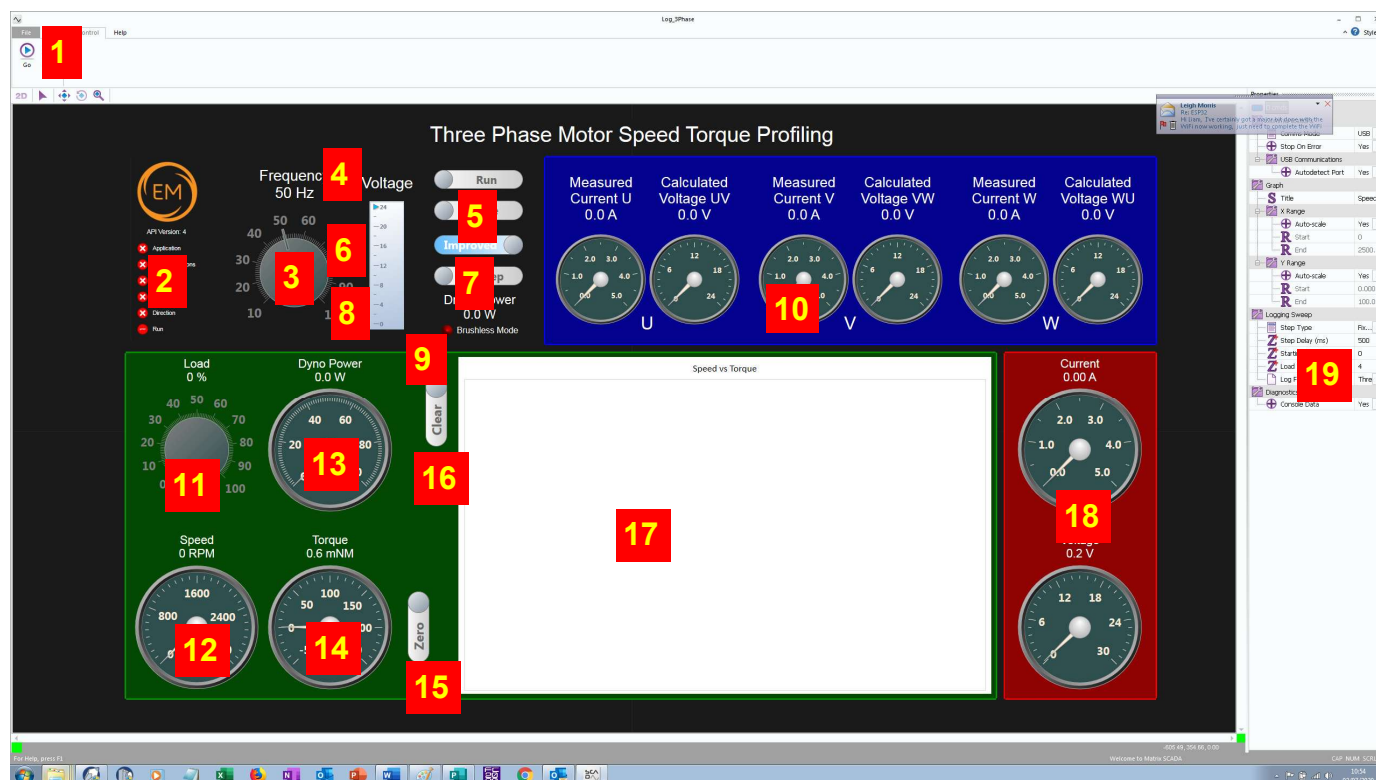
- 11) Dynamometer speed in RPM
- 12) Dynamometer power in watts
- 13) Torque in mNm
- 14) ZERO: Zeros the torque sensor
- 15) REAL/EFFECTIVE: adjusts the way the graph plots the output waveform.
- 16) PHASE / TIME: shows X axis in degrees or in mS
- 17) CAPTURE: captures a waveform.
- 18) Oscilloscope plot area
- 19) Internal ammeter and voltmeter V1 and I1
- 20) Properties

This application runs the three phase motor and the Brushless DC motor.

# Reference

## Three phase AC automatic

# Modern electrical machines



The file name for this program is 'Open Log\_3Phase.bat'

- 1) Program control - hit 'PLAY' to start
- 2) Flags for App running, USB comms and Motor in place.
- 3) Frequency of output
- 4) RUN: needs pressing to start the program
- 5) SINE / DIGITAL: Sine or Square wave output
- 6) 20V / 12V output: alters the output voltage
- 7) IMPROVED / STANDARD: alters output wave-form quality
- 8) SWEEP: starts sweep function
- 9) BRUSHLESS MODE: indicates Brushless motor is plugged in
- 10) Meters for the output - RMS values
- 11) Dynamometer load as a %

- 12) Dynamometer speed in RPM
- 13) Dynamometer power in watts
- 14) Torque in mNm
- 15) ZERO: Zeros the torque sensor
- 16) Graph / log clear button
- 17) On screen graph of plot data
- 18) Internal ammeter and voltmeter V1 and I1
- 19) Properties

This application runs the three phase motor and the Brushless DC motor.

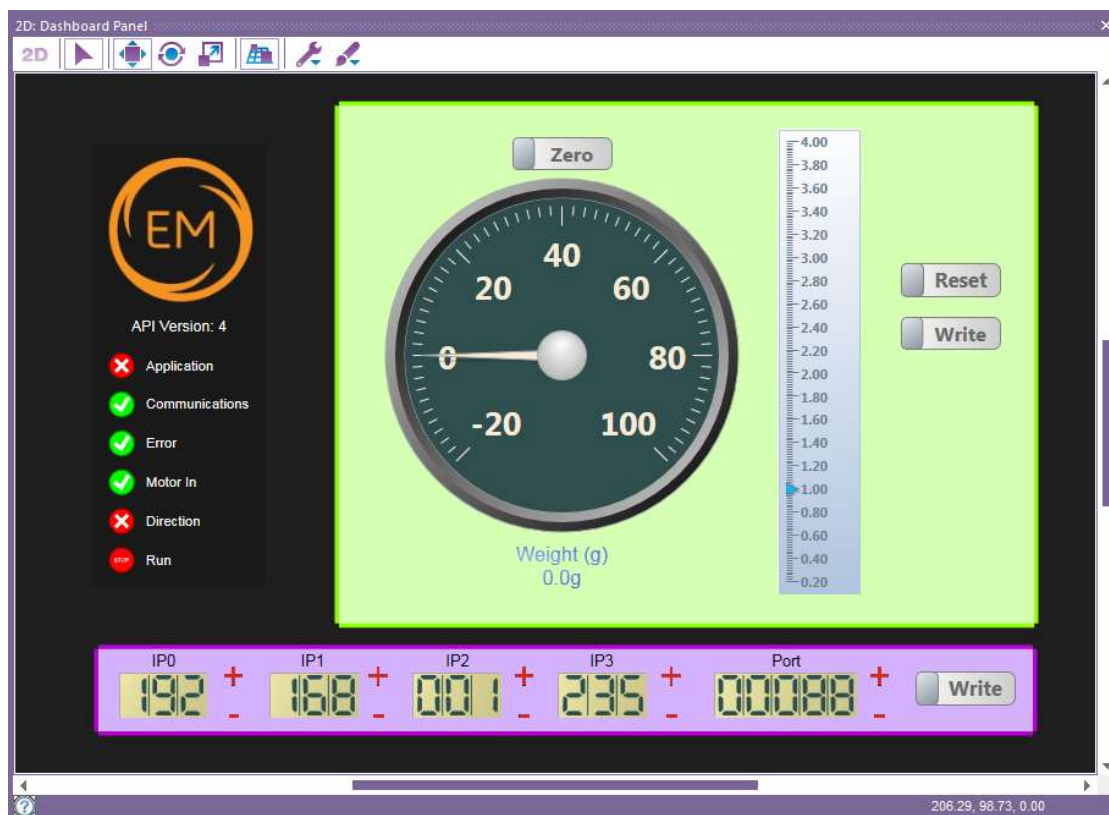
When the SWEEP button is pressed the control box will - at the specified frequency output take readings from 0 load to 100% load in 2% increments. The results will be stored in the file LOG\_3phase.CSV for plotting in Excel.

Up to 8 data sets can be captured and displayed at one time.

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## Calibration and IP set up

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The file name for this program is 'Open Calibration.bat'. This software application has two functions: it allows you to calibrate the load cell and it allows you to set the IP address of the control unit.

### Calibration

- 1) Connect the Dynamometer to the control unit with the 25 way D type.
- 2) Twist the dynamometer anti clockwise so that it is not pressing on the load cell at the back of the Dynamometer.
- 3) Press PLAY on the software App. Press RESET. Press Zero.
- 4) Place a 20 gram weight on the end of the load cell. Use the slider multiplier on the App to adjust the dial so that it reads 20g.
- 5) Click on the Write button to save the calibration into the control box.

tion into the control box.

- 6) Remove the weight. Press Zero. Your Control unit and Dynamometer are now calibrated together.
- 7) Check the calibration by putting the weight on and making sure the display reads 20g.

If you have multiple sets of Electrical Machines then make sure that they are paired for the next time you use them. This will have been performed for you in the factory so this should not need doing on starting.

### IP set up

- 1) Press PLAY on the App
- 2) Use the '+' and '-' buttons on the IP number fields to set the IP address you want
- 3) Click on the WRITE button to save this IP address to the Control box.

## Remote Lab Connection

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This document outlines the process users should follow to remotely connect an electrical machines control box to a network or internet to allow remote control. This can only be actioned on units purchased since June 2020.

### Accessing using local network

The default IP address of the Electrical Machines box is 192.168.1.235 and the default port is 88. The IP address and Port can be changed by connecting to the box using a USB connection and running the calibration application. The Ethernet connection must be connected to the network and then the IP address and Port can be used in the software applications to connect to the Electrical Machines box.

### Accessing using the Internet

Before accessing the Electrical Machines box from the internet make sure the box is functioning correctly on the local network. This helps to pin point any connection problems down to the IP and Port of the box or the configuration of the router.

To access the Electrical Machines box from the Internet the user must ensure they have enabled port forwarding on their own router, (the connection from your local network to the internet). This takes incoming connections from the internet and routes them to the correct internal IP address. Use the routers IP address in the software applications.

Information on enabling Port Forwarding on your router: <https://portforward.com/>

Information on finding your routers IP address. <https://whatismyipaddress.com/>

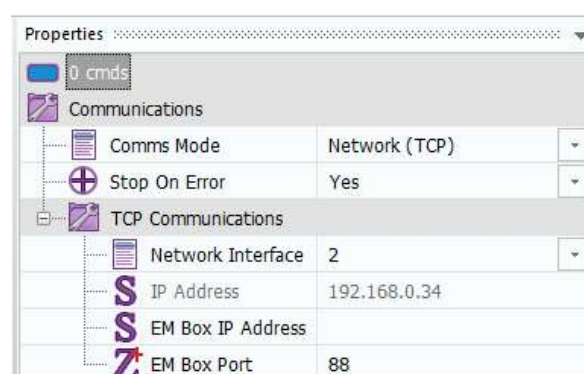
On a home routers (for example in the students' or lecturers home) you may also need to enable a demilitarised zone (DMZ) to the Electrical Machine box IP address to allow incoming Internet traffic to reach it. A commercial router may not have to do this.

### Configuring the software application

Using the software application downloadable from the Matrix website: [www.matrixtsl.com/electrical-machines/resources/](http://www.matrixtsl.com/electrical-machines/resources/), the user starts the Electrical Machines software application for the specific machine they have in test.

Ensure Network (TCP) communications is selected (not USB). Under the TCP communications menu, select the network interface that provides a sensible IP value. In the EM box IP address, enter the IP for your control box. In the EM box Port, enter the Port number for your control box.

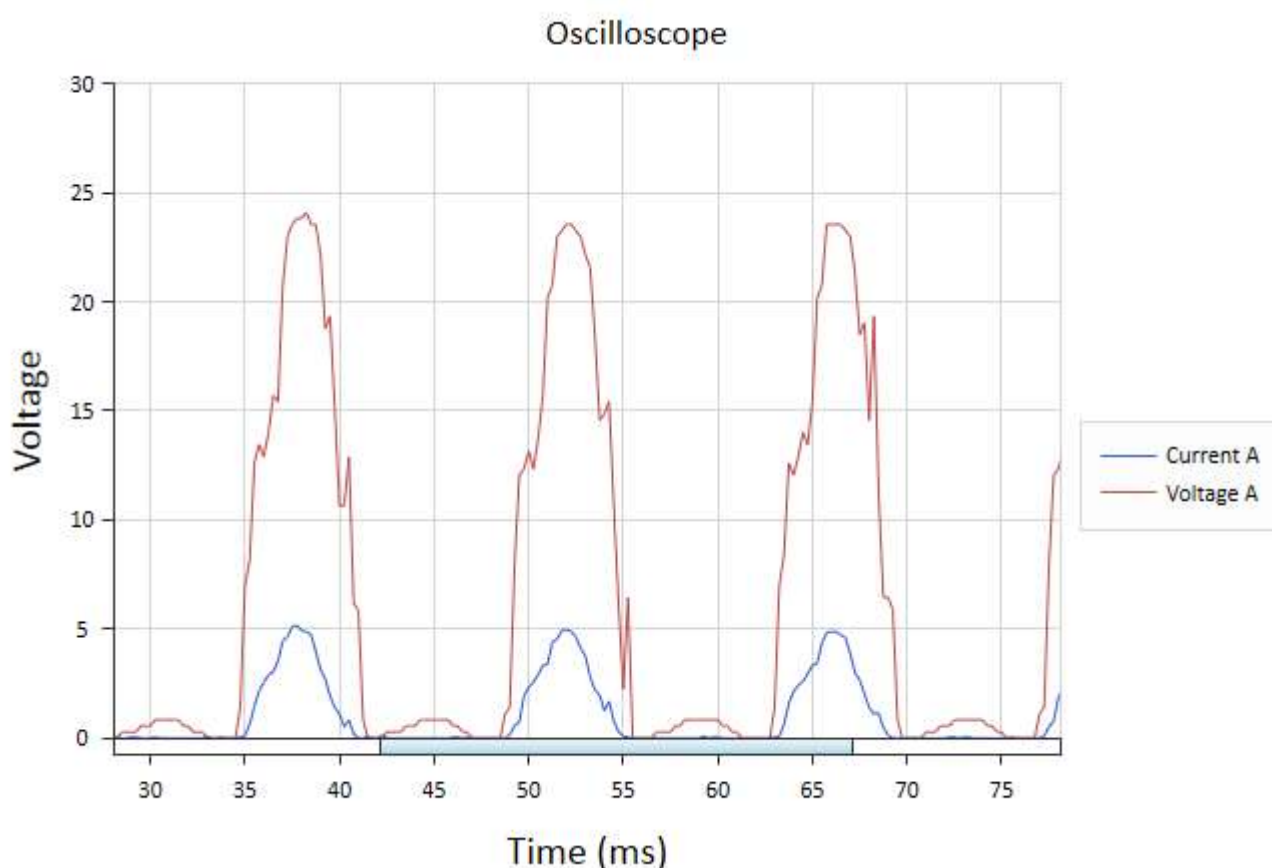
*NB: Matrix TSL are unable to provide help in configuring the IP systems of third parties because of the complexities inherent in this. Local area network managers should be able to provide this support.*



# Reference

## The on- board oscilloscope

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The oscilloscope in the Control unit takes measurements of every ammeter and voltmeter in the unit at a rate of 64kHz. There are 16 channels in total - that gives us an effective measurement rate of 4kHz for each channel.

Y axis, Timebase and traces can be set by clicking on the appropriate words in the Properties section at the right hand side of each application.

To alter the time base: click on the oscilloscope area and use the mouse wheel to zoom in and out.

The oscilloscope is useful for seeing voltages and currents at close to the driving frequencies. In fact we use a small capacitor on the input to deliberately slow the effective rate of

sampling to make the instrument work better for this purpose.

Because the PWM frequency used is 7kHz and the effective sampling rate is 4kHz then Nyquist effects can be seen when zooming in to examine the PWM drive frequency. Because of this an external oscilloscope needs to be used when examining waveform structure in detail.

The oscilloscope is particularly good at showing currents the pattern of voltage and current and the phases between voltage and current.

The oscilloscope in the three phase programs has the ability to show Real voltage or Effective voltage as standard. Users at this level should always use Effective voltage.



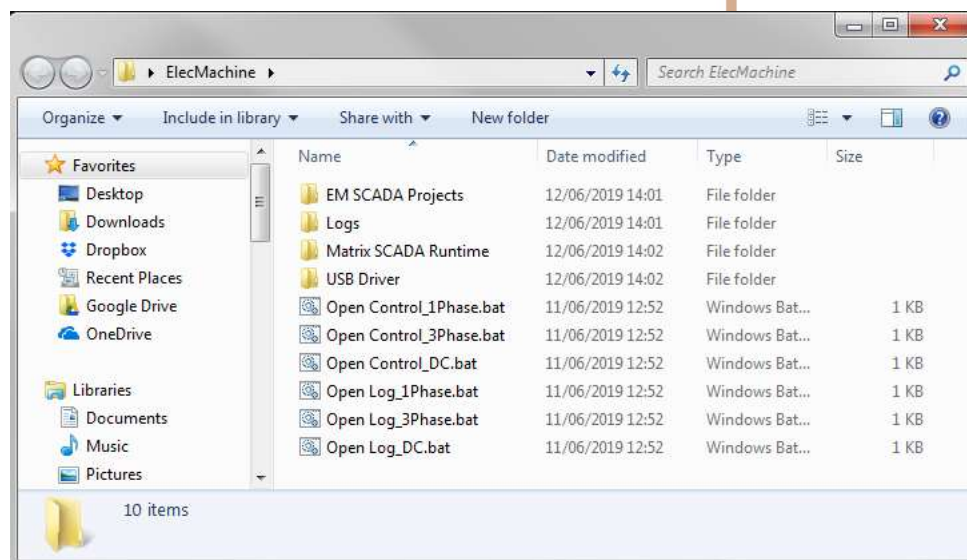
# Reference

## Software installation

## Modern electrical machines

### Copying over the software

The software for Electrical Machines is available as a download from the Matrix web site. When you unpack the zip file you will see the following files:



There are 6 applications that are used for the Electrical Machines range. These are documented below.

Please copy these programs and directories into a suitable location on your hard drive. You can make short cuts to the \*.BAT files if you wish.

Run each program by clicking on the '\*.BAT' file.

### Installing drivers

Go into the USB drivers folder and run the 32 bit or 64 bit as appropriate for your computer. This should install the driver for the control box.

You can check the driver is installed properly by looking at Device Manager and checking that the Electrical Machines controller appears under Ports - see opposite.

